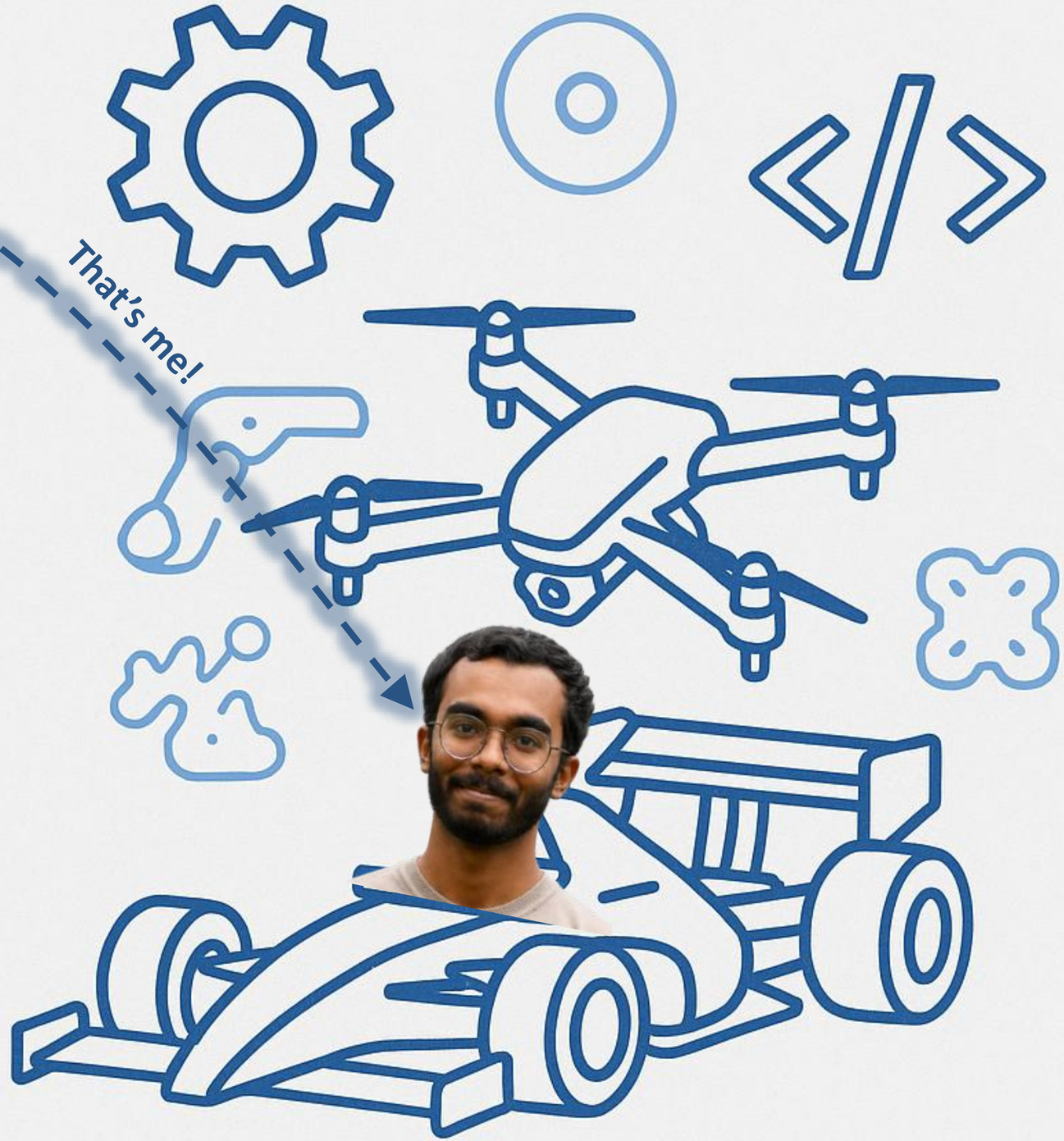
Srivisweswara Mohan Santhi

PORTFOLIO OF PROJECTS AND ACHIEVEMENTS



Engineer | Innovator | Developer

vichu26092003@gmail.com

https://www.linkedin.com/in/srivisweswara-mohan-santhi-2a2461233/

https://srivisweswara.com/

Overview

- **Purpose of this deck:**

  I am highly passionate about Electric Vehicles and Embedded Systems. During my studies, I have been a technical lead for several teams that have won national level competitions and received acclaim from various organizations in India.

  This deck serves to highlight my technical accomplishments.

- **About Me**

  2026 MSc Embedded Systems Engineering from University of Leeds.

  2025 B.E. Electrical and Electronics Engineering from Sri Krishna College of Engineering and Technology.

  Work mainly with ARM-based processors and focus on building and programming hardware.

  Proficient in Embedded C, C++, Rust (Intermediate), Assembly (Intermediate) and Data Structures & Algorithms.

  Hands-on knowledge drone and Electric Vehicle building, PID tuning.

  Lead technical projects and working with teams to turn ideas into real, working systems.

  2024 & 2025 National level EV competition (EKVC) Champions

  2023 SIH winner for national level hackathon for design and build of a drone

- **Projects highlighted in this slide deck**

  Embedded Systems Projects

  Electric Go-Kart

  Drone

  Websites

Cortex-A9 Bare-Metal DSP System

- **Project Focus**

•**Platform:** DE1-SoC, Cortex-A9, Cyclone V, audio and display peripherals

•**What I learned:** system-level bring-up, DSP optimisation, and getting several low-level blocks to work together cleanly

•**Why it matters:** I can help a company debug hard embedded problems, tune bottlenecks, and integrate hardware into a reliable product

- **Overview**

•Built a bare-metal DSP platform on the DE1-SoC that runs audio, SDR, sensor, and image-processing pipelines without an operating system.

- **Highlights**

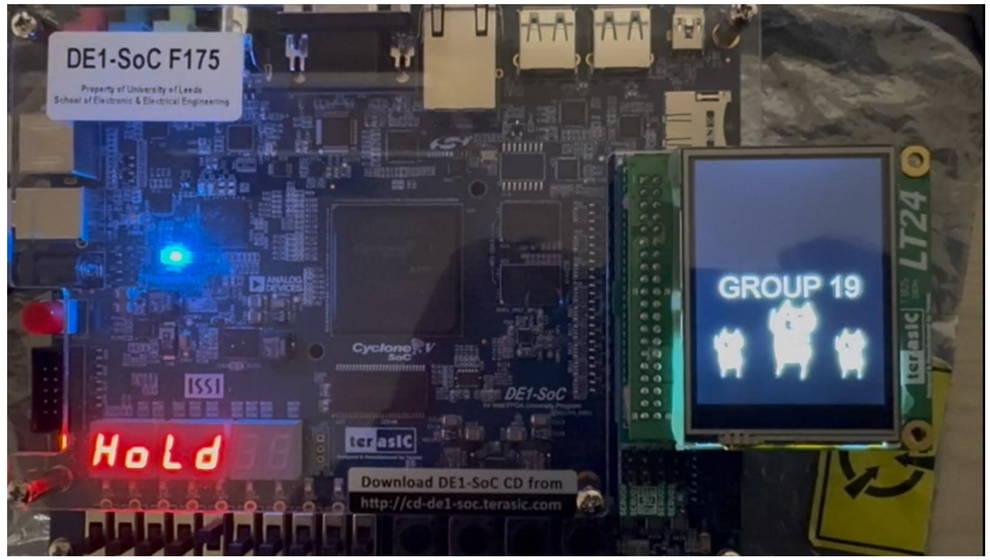
•**Performance:** Used ARM NEON to reduce FIR and FFT runtimes with clear, measured gains.

•**Integration:** Brought UART, codec, LCD, storage, and board controls into one working runtime.

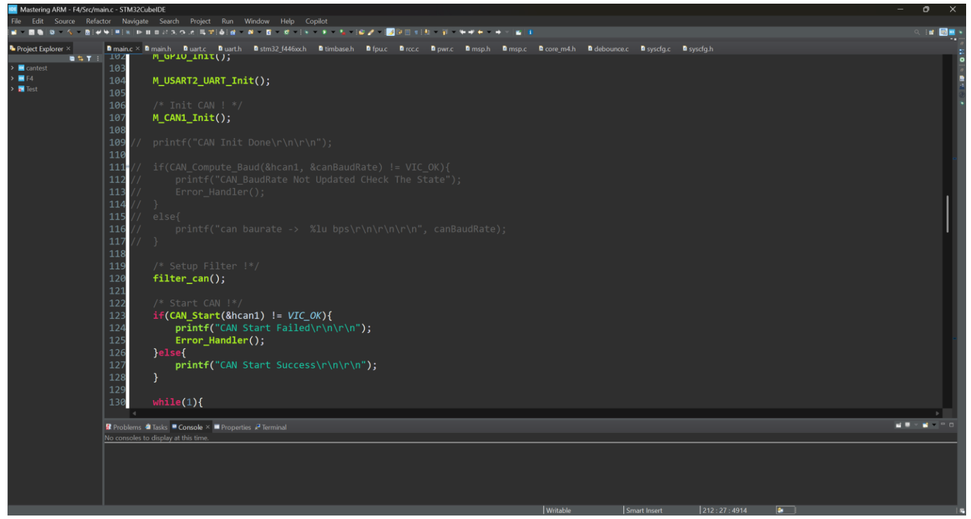
•**Company value:** This is the kind of work I can bring when a team needs low-level ownership, profiling, and dependable embedded integration.

- **Links**

•Project Page | GitHub



Bare-Metal STM32F4 Library



- **Project Focus**

•**Core:** STM32F4, startup code, linker scripts, and reusable low-level drivers

•**What I learned:** how to control boot flow, memory layout, and peripheral behaviour instead of treating them as black boxes

•**Why it matters:** I can step into low-level firmware work, debug with confidence, and build foundations that other engineers can extend

- **Overview**

•Built a from-scratch STM32F4 firmware base to own the startup flow, memory map, and register-level drivers.

- **Highlights**

•**Startup:** Worked directly with linker scripts, reset flow, and hardware initialisation.

•**Drivers:** Kept the code close to the registers so behaviour stays predictable during debugging.

•**Company value:** Useful for products that need solid board bring-up, reusable BSP work, and firmware that is not dependent on generated code.

- **Links**

•Project Page | GitHub

Encryption and Cryptography Library

- **Project Focus**

•**Core:** encryption workflows, reusable interfaces, and secure data handling

•**What I learned:** security work needs the same discipline as firmware work - clear interfaces, traceable behaviour, and careful data movement

•**Why it matters:** I can bring a more security-aware mindset to embedded and low-level software teams

- **Overview**

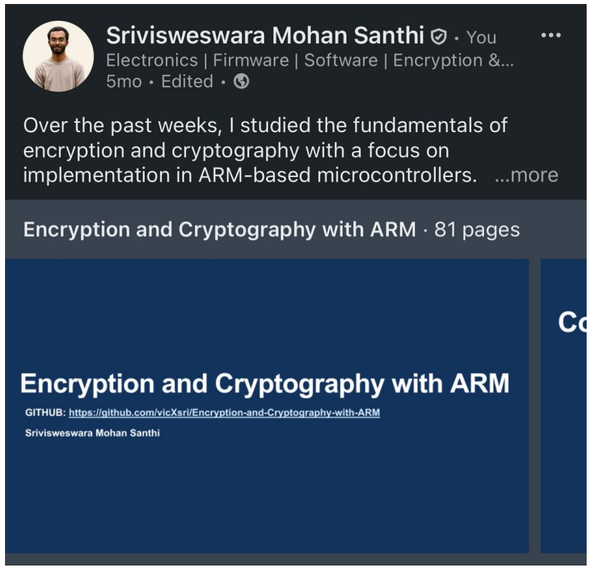
•Built this library to understand how encryption code should be structured, reused, and tested rather than treated as a one-off demo.

- **Highlights**

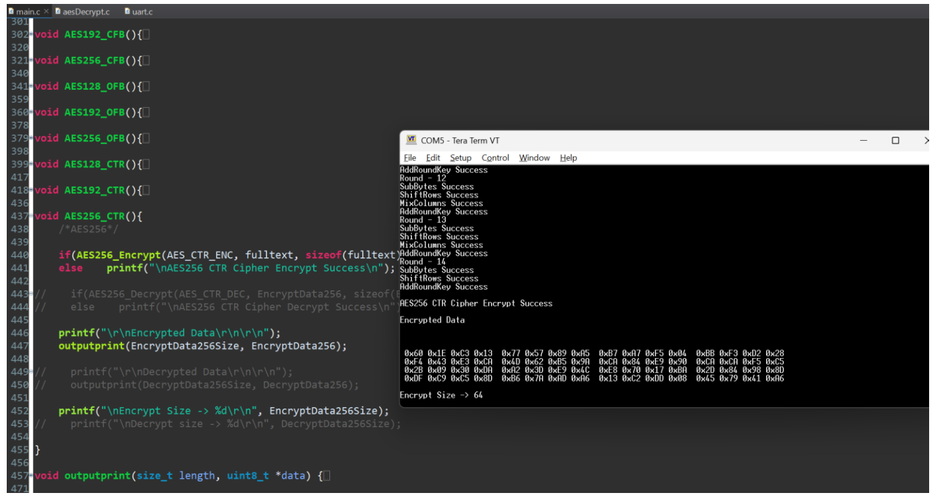
•**Design:** Focused on clean APIs and predictable handling of protected data.

•**Engineering:** Used the project to strengthen algorithm-level thinking beyond firmware alone.

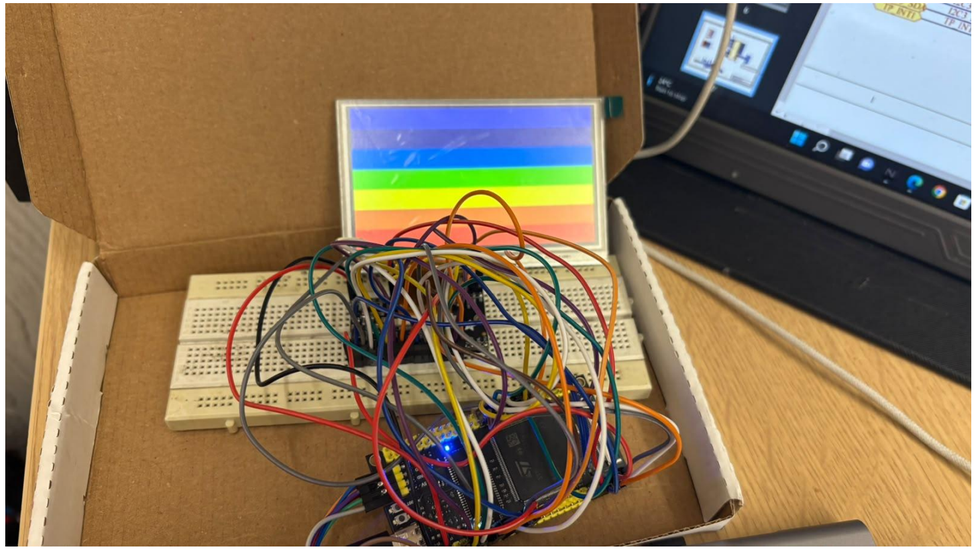
•**Company value:** Useful for teams building products where correctness, trust, and clean software structure matter.



Links: Project Page | GitHub Profile



Retro Gaming Console



- **Current Build**

•**Platform:** STM32H7 handheld console build with LTDC display and custom firmware

•**What I learned:** staged bring-up, UI integration, and steady hardware debugging on a live build

•**Why it matters:** I can help teams move an early prototype from messy hardware tests to a usable product path

- **Overview**

•This project is teaching me to build the firmware stack step by step, from display and I2C bring-up to LVGL-based UI work.

- **Highlights**

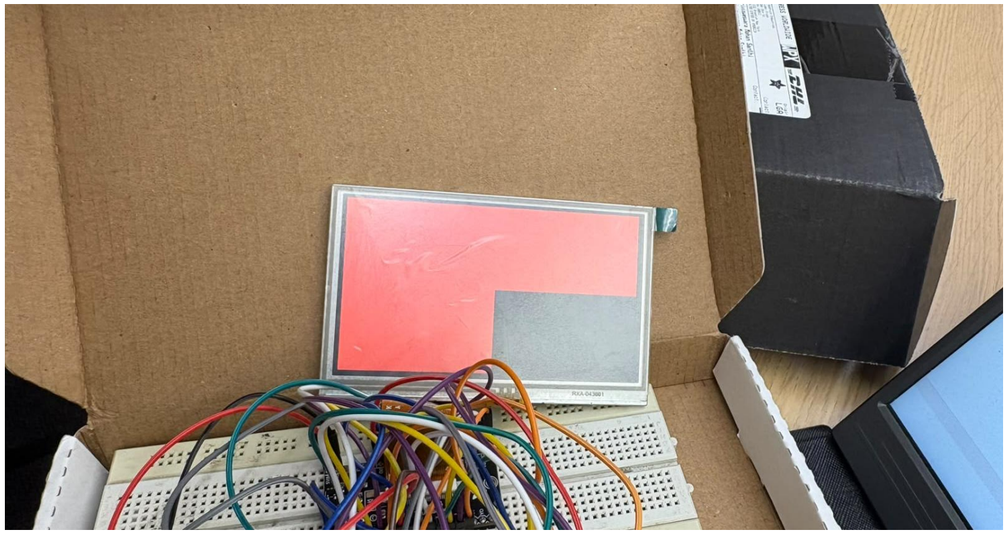
•**Bring-up:** Worked through LTDC, RCC, I2C, and display testing as the hardware came together.

•**UI Work:** Integrated LVGL and started rendering live data on the screen.

•**Company value:** Useful for products that need patient debugging, hardware-software coordination, and clear technical documentation.

- **Links**

•Build Journal



Electric Go-Kart – Gandiva Racing

Links: Project | GitHub

- **2022-2023: Chief Electrical Officer and Innovation Lead**
  - **Summary:** 1st season. Failed technical inspection. Finished last. Lots of learning.
  - Founding member of team to participate in national level EV competition to build 6 kW electric go-kart with 60V 150Ah NMC battery
  - Introduced F1 style steering wheel with custom instrument cluster
- **2023-2024: Chief Electrical Officer and Innovation Lead**
  - **Summary:** iterative improvements. Budget constrained. Solid technical progress.
  - Improved harness, reduced weight, reached 80km/hr.
  - Won EKVC 2024 championship
- **2024-2025: Promoted to Vice captain**
  - **Summary:** Our crowning achievement. My technical masterpiece ☺
  - Simplify complexity, Reduced weight to <120kg, Aerodynamic design, 104km/hr, custom 72V 40Ah HV battery pack with Molicel cells.
  - Won EKVC 2025 championship and placed All-India #3 at GKDC 2025
- **My specific contributions and technical details:**

  - **Electrical:**
    - Designed complete EV schematics in ACAD electrical. Designed and built a single integrated wiring harness
    - Used 25mm^2 silicone cables for high-current (up to 125 A at 72 V, peak 8.5 kW). 0.5 mm^2 for control lines, 0.25 mm^2 for signal-level electronics
    - Used 90-series waterproof connectors and automotive-grade insulation tape
  - **Control Systems and PCB design:**
    - **PCB Design:** Altium Designer. Created a custom steering wheel PCB for control and feedback
    - **Microcontroller:** STM32F446RE. Developed embedded firmware using STM32CubeIDE
    - **Protocols Used:**
      - **CAN 2.0B**: For inter-module communication (BMS, display, logger)
      - Wrote firmware to manage request/response format (BMS) and continuous broadcast (MCU)
      - Used TJA1050 CAN transceiver
      - **SPI:** For WS182b LED strip control and SD card data logging
      - **UART:** For communication with the DWIN TFT display
  - **Firmware**
    - Managed CAN message optimization to multiplex BMS + MCU data
    - Programmed speed-responsive LED animations for the WS182b strip
    - Wrote SPI and UART drivers for modular communication
    - Designed GUI and boot animation using DWIN software tools
  - **Battery Pack Engineering**
    - Battery Configuration: 72V, 40Ah (Molicel cells)
      - Designed the pack architecture and validated sizing with MATLAB
    - Protection Devices:
      - 125 A MCCB (C-Rated) for HV safety
      - 250 A contactor for ignition and high-voltage line control
      - Simulation & Validation
  - **MATLAB Simulink:**
    - Ran simulations using the Gandiva Drive Cycle to determine Wh/km and total Ah requirement
  - **SolidWorks Flow Simulation:**
    - Validated aerodynamic CAD model. Verified airflow around body panels and cooling efficiency for motor and battery
    - Informed drag reduction and optimal component placement

Electric Go-Kart – Gandiva Racing – Project details


Instrument cluster


Light strip

- **LED light strip:** Designed an F1 style light strip on top of the steering wheel

  Problem statement: During races, kart drivers could not easily read instrument cluster, leading to poor speed awareness.

  Solution: I am very proud that this is my personal innovation, using a WS182b LED strip at the top of the steering wheel. LED lighting was controllable per track speed limits. Got good driver feedback!

- **Steering instrument cluster:**

  Introduced first-of-its kind instrument cluster in the league. Drivers got live status updates like battery temp, SOC percentage, controller temps, motor temps, RPMs, Speed, error codes, etc.

- **Kart Aerodynamics:** Reach a top speed of **104 km/h using just 5.5 kW**

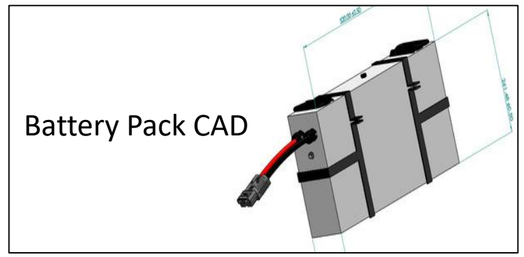
  Reduced the **drag coefficient (Cd)** and **frontal area (Af)** of the kart using light weight full-body shell and aero elements in CAD, keeping airflow and packaging in mind.

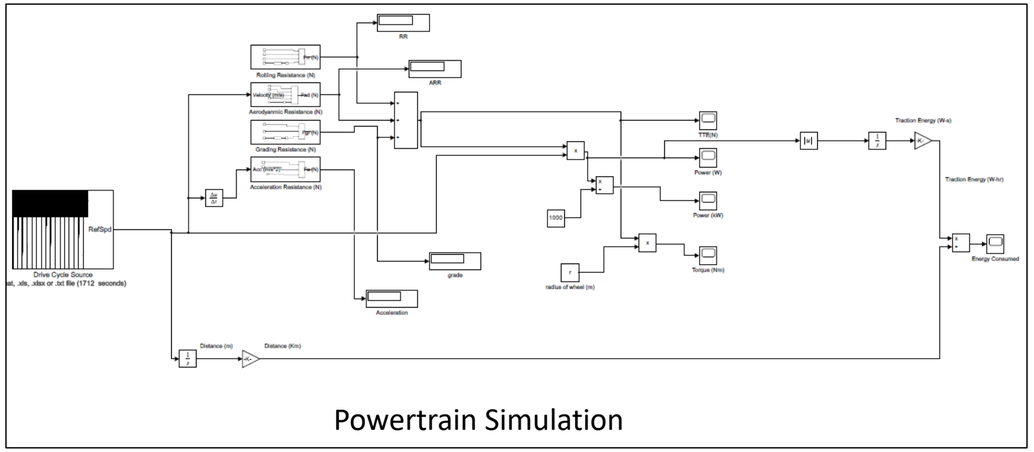
  Methodically optimized parameters like **Cd** (Co – Efficient of Drag), **µrr** (Co – Efficient of Rolling Resistance) and many others by testing & **SolidWorks Flow Simulation** to simulate different designs for aero elements. This helped us identify drag-heavy zones and improve the design.

  Very proud that I taught myself aerodynamic concepts to contribute to optimally solve these problems

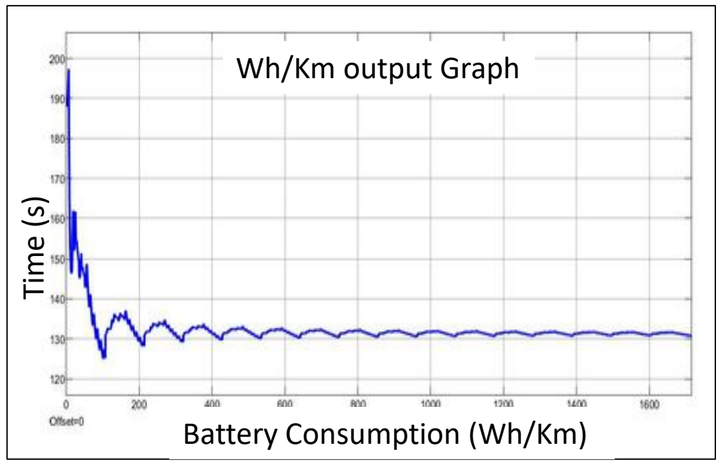
- **Light weighting:**

  Using aerodynamics simulations, retrieved the wh/km, and used it in a battery pack simulation to optimize and build a **very light weight battery pack**.

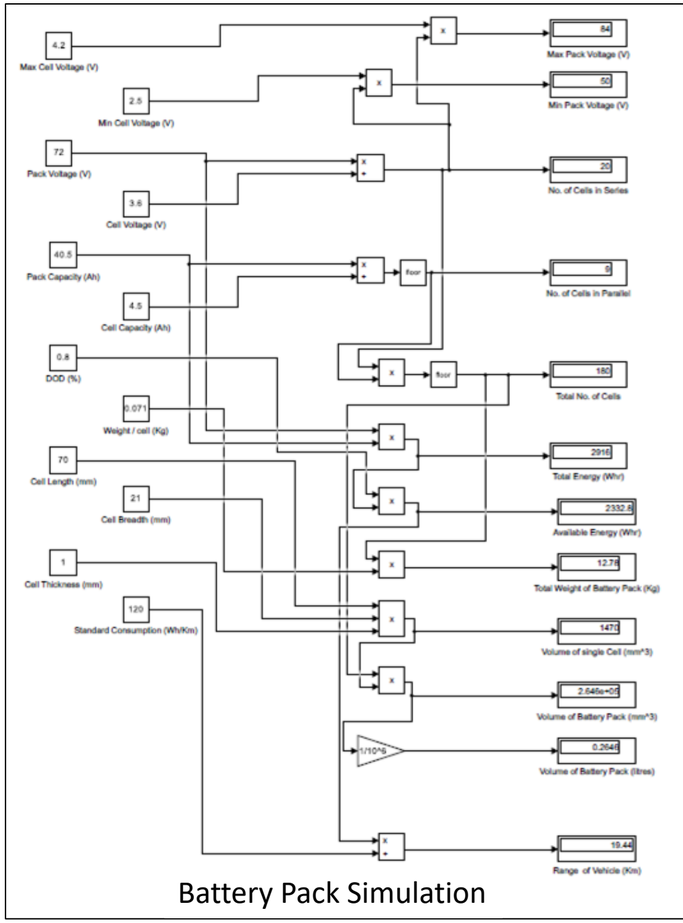




Powertrain Simulation

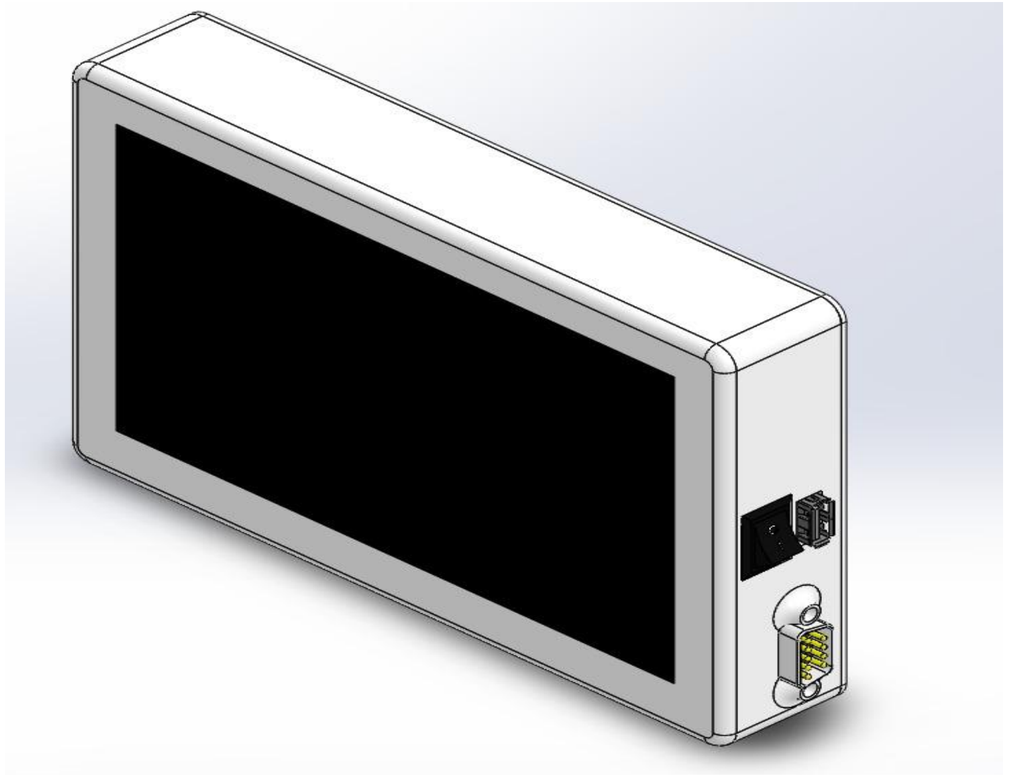


Battery Consumption (Wh/Km)



Battery Pack Simulation

GCAN – CAN Analyzer and Data Logger



Tech Stack

- **Hardware:** Raspberry Pi, STM32, MCP2515 CAN, Waveshare HDMI
- **Software:** Python, PySide6, SocketCAN, Linux
- **Protocols:** CAN Bus, SPI

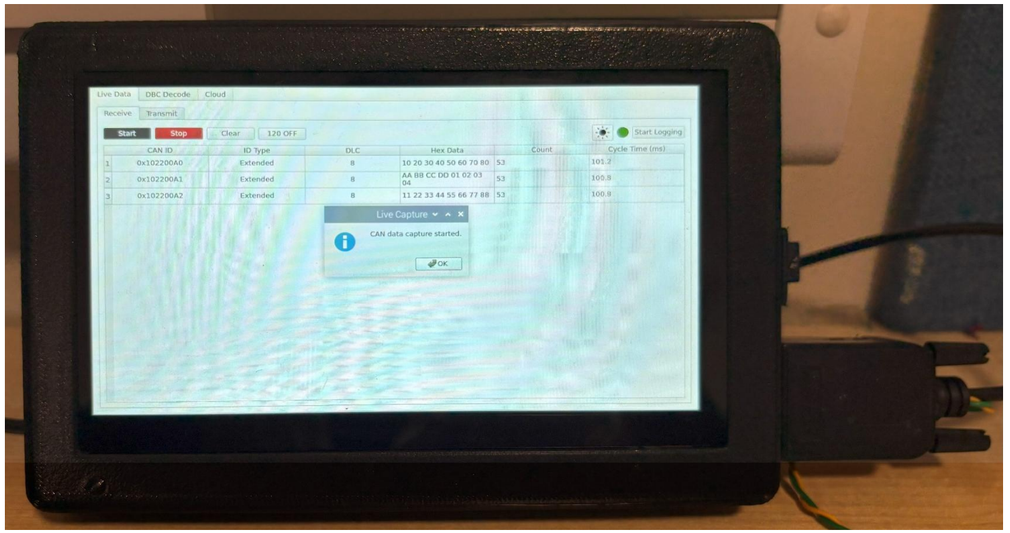
Overview

- Built a **real-time CAN bus data logger** using Raspberry Pi with a custom **Python GUI** for monitoring, decoding, and storing CAN data.

Highlights

- **GUI:** Multi-tab PySide6 interface with start/stop logging, error handling, and CSV export.
- **CAN Integration:** Configured MCP2515 + SocketCAN.
- **Data Handling:** Real-time logging to CSV/SQLite with tabular & list views.
- **Enhancements:** Display brightness control, modular design for future cloud/analytics features.

Link: Project Page



Drone Project

Link: Project Page

- HAWK v1.0
  Problem statement: Geo-tagging of plantation in the catchment area of hydro project:

- As part of **Smart India Hackathon 2023**, our team built an autonomous drone system for the problem statement: *"Geo-tagging of plantation in the catchment area of hydro project"*, given by the Ministry of Power. The goal was to track plantation growth and detect early signs of plant stress or disease using aerial monitoring.

- The drone platform included a **Jetson Nano**, which ran a machine learning model in Microsoft Azure to detect crop disease in real time using data from **RGB and NIR cameras**. The system also collected **GPS-tagged images**, enabling us to map the plantation area and analyze vegetation health using **NDVI and EVI** indices.

- I was responsible for assembling the drone, tuning the **PID controllers** in Betaflight for stable flight, and programming the drone's firmware. I also worked on integrating the ML model with the Jetson Nano, and helped sync image data with the GPS module for accurate geo-tagging.

- After flight missions, we used **ArcGIS Pro** to process and visualize the data. Our solution was declared the winner at SIH 2023, and we were awarded a **₹1,00,000 cash prize** by the Ministry of Power for building a practical, field-ready system.

Website Development

- E-Kart Club Website

  Built for our college's **Electric Kart Club** to showcase team updates, projects, and events

  Collaborated with a teammate; focused on frontend development and GUI styling

  Tech Stack Used:

  → React
  → Tailwind CSS
  → Firebase (hosting & backend)

  My Contributions:

  - Wrote and styled the entire **frontend interface** using Tailwind CSS
  - Designed the **GUI layout** and component structure
  - Maintained and deployed updates to Firebase
  - Ensured full responsiveness and performance optimization

Link : Portfolio | E-Kart

- Garden Website – Freelance Project :

  Designed for a **local garden services business** to create an online presence

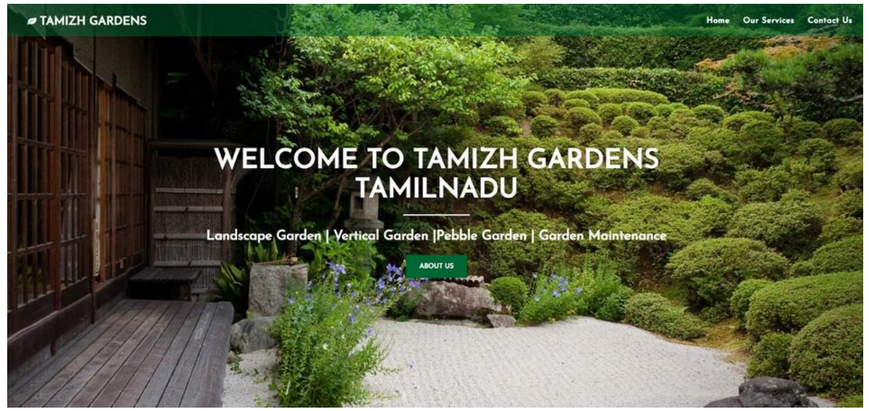
  Focused on simplicity, fast loading, and mobile responsiveness

  Tech Stack Used:

  → HTML
  → CSS
  → Firebase (for hosting)

  My Contributions:

  - Developed the full static site from scratch
  - Handled all **layout, styling, and deployment**
  - Worked closely with the client to reflect their brand and services



Link : Garden

The value I bring to the table

What I Bring to the Table

- **Hands-on embedded ownership:** Comfortable with schematics, board bring-up, low-level C, and real hardware debugging.
- **Reusable firmware thinking:** I build drivers, startup code, communication stacks, and tools that other engineers can extend.
- **Systems integration:** I connect hardware, firmware, UI, logging, and testing so the full product works together.
- **Measured problem solving:** I use profiling, simulation, and structured debugging to improve performance and reliability.
- **Team value:** I document clearly, collaborate well, and stay useful across electrical, firmware, and product work.